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Aerogel-based Leidenfrost vapor percolator for ultra-fast thermal cooling

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The Leidenfrost effect—where liquid droplets levitate on superheated surfaces—hinders efficient heat transfer. Conventional mitigation strategies typically rely on rigid materials with complex surface engineering, but their applicability is limited by stringent weight requirements, especially in aerospace systems where every gram counts. Here, through controlled tuning of the drying stress field and targeted welding of hollow nanofibers, we construct an aerogel-based Leidenfrost vapor percolator featuring directed nano/micro tunnels for efficient vapor evacuation and flow-topology reconstruction. This ultralight (7 mg cm^{-3}) permeator robustly suppresses the Leidenfrost effect up to 1000 °C and achieves a high heat flux of 110.43 W cm^{-2}. Moreover, the permeator delivers a temperature drop of 760 °C under multi-flow injection, corresponding to a density-normalized Leidenfrost temperature reaching 142.9 °C cm^3 mg^{-1}. This two-order-of-magnitude enhancement over conventional materials highlights the pivotal roles of hierarchical structuring and vapor-channel regulation in overcoming the Leidenfrost barrier.

Effective cooling of overheated surfaces is of paramount importance in diverse fields, from everyday technologies to high-demand industrial sectors such as steel manufacturing, aerospace, and nuclear power generation[1–7]. Active cooling methods, including spray and jet cooling, are highly favored due to their efficiency and reliability. However, their performances are often limited under extreme temperatures. When surface temperatures exceed the Leidenfrost temperature (LFT, typically 250 °C for water in metal surface), a continuous vapor film (λ_{vapor} = 0.026 W m^{-1} K^{-1} @100 °C) forms, leading to thin film boiling, which increases thermal resistance and diminishes heat transfer efficiency[2,3,8]. Therefore, the phenomenon of interest primarily manifests within the thermal regime spans from the boiling point of water (100 °C at 1 atm) to the LFT, implying that elevating the LFT is a key strategic approach to sustaining efficient thermal transport[3,9].

Current research for improving the LFT has primarily focused on rigid materials combined with complex surface engineering techniques[2,6,9], including thermally conductive micropillars[7,10–12], tailored surface roughness[13–15], and porous structures[16]. These designs enhance solid-liquid interfacial contact and facilitate phase change[10,17,18], while the incorporation of thermally conductive surface textures further reduces interfacial thermal resistance, thereby elevating the LFT to approximately 453–570 °C[8,17]. Nevertheless, most of these techniques rely on dense substrates with high intrinsic mass and limited structural adaptability, which constrain their applicability in weight-sensitive and thermally extreme environments such as spacecraft thermal protection systems. The challenge, therefore, is to design a lightweight and mechanically compliant cooling architecture that concurrently mitigates the Leidenfrost effect and fulfills the stringent mass requirements of advanced thermal protection systems.

Ultra-low-density ceramic nanofiber aerogels are promising for high-temperature thermal protection due to their high porosity, excellent chemical and thermal stability, and pronounced elasticity[19–21]. Critically, the highly cross-linked three-dimensional (3D) nanofibrous porous network in these aerogels provides optimized pathways for heat and mass transfer while maintaining structural integrity under extreme conditions[19,20,22]. Building on these features,

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we propose an LFT-enhancement design: a ceramic nanofiber aerogel-based vapor permeator (NAVP), in which the nanofibers are hollow, thereby facilitating the efficient removal of vapor induced by the Leidenfrost effect and elevating the LFT. Nevertheless, the proposed design presents considerable challenges, particularly in ensuring the structural integrity of the nanofiber tunnels during the drying process of hollow ceramic nanofibers (HCNFs) fabrication and in avoiding pore blockages during aerogel preparation.

To overcome these challenges, we present a compositional-structural engineering strategy that integrates solution exchange during HCNF preparation with congener-targeted welding of the resulting aerogels, thereby facilitating the advancement of ultralight vapor permeator. By leveraging the cavitation effect of low-surface-energy solvents to alleviate drying stress and employing confined welding to optimize interfacial coupling—crucial for preventing excessive crosslinking within the aerogel network—our NAVPs exhibit integrated nanoscale (nanofiber pores) and microscale (aerogel pores) tunnel architectures that facilitate efficient vapor evacuation during the Leidenfrost effect. Consequently, the ultralight (7 mg cm^{-3}) NAVPs effectively suppress the Leidenfrost effect at temperatures up to 1000 °C with a heat flux of 110.43 W cm^{-2}, performing comparably to the best-in-class thermal cooling materials[3]. Moreover, they can reduce the surface temperatures to about 240 °C ($T_{initial}$ = 1000 °C, ΔT = 760 °C) under a cumulative water droplet cooling rate of 9.5 mL min^{-1}. More importantly, unlike traditional cooling systems that rely on structural stability, the failure threshold of NAVP materials is determined by the intrinsic melting point of the ceramic nanofibers, marking the conceptual shift in thermal management technology from structural defense to material intrinsic defense.

Results

Design and principle of aerogel-based Leidenfrost vapor percolator

Thermal cooling technology in extreme environments, such as advanced electronics cooling, high-performance computing systems, and aerospace applications, requires cooling materials capable of withstanding heat fluxes that often exceed 100 W cm^{-2}[3,23]. In this context, phase-change cooling techniques attract considerable attention due to their exceptionally high heat flux (q) and heat transfer coefficients (HTC) (Fig. 1a)[23–25]. The heat transfer curve of phase-change cooling is typically depicted in Fig. 1b[9,26]. An appeared trend is observed where the q increases with the super heat T_w, particularly in the nucleate boiling regime (regions ii in Fig. 1b). However, once the T_w surpasses the LFT, the Leidenfrost effect occurs due to the film boiling (Regions iii and iv in Fig. 1b), causing a rapid decline in q and often resulting in device burnout or catastrophic system failure[2,9,26].

These results were further elucidated using the experimental setup shown in Supplementary Fig. 1, designed to capture the dynamic behaviors of water droplets on extremely superheated surfaces (T_w = 1000 °C). Notably, the droplets (~16 µL) were deposited in a controlled quasi-static manner (velocity of ~18 cm s^{-1}), which corresponds to the standard conditions for the defining LFT, rather than being subjected to high-impact dynamic contact. Concretely, upon impact with a superheated bare metal surface (SUS 304 stainless steel, surface roughness Ra<0.1 µm), the droplet rapidly rebounded, remaining within the Leidenfrost regime throughout its duration (τ) of around 41 s due to the stable vapor layer characteristic of the Leidenfrost regime (Fig. 1c). Distinctly, the developed NAVP surface, characterized by honeycomb-like micro networks with aligned pore walls, strengthened capillary action of liquids and accelerated the evacuation of vapors (Fig. 1d). Thus, the droplet lifetime τ was drastically reduced to 0.417 s—approximately 98 times shorter than that on the bare metal surface, demonstrating a pronounced suppression of the Leidenfrost effect and an improvement in cooling performance. Notably, these phenomena were similarly demonstrated with liquid nitrogen, whose boiling point is about 300 °C lower than that of water. As shown in the Supplementary Fig. 2, the liquid nitrogen spread evenly over the NAVPs at 1100 °C, resulting in a sub-second lifetime. By contrast, when liquid nitrogen droplets impact bare metal surfaces, they exhibit vigorous bouncing, incomplete wetting, and persistent vapor films, consistent with the classical Leidenfrost phenomenon (Supplementary Fig. 3).

3D laser scanning microscopy was employed to characterize the surface microstructure of NAVP. As shown in Fig. 1e, the surface exhibits a randomly distributed rough morphology, with an arithmetic average height (Sa) of 17.58 µm. More crucially, the distinctive surface morphology of the NAVP establishes a strong coupling with the escaping vapor flow, thereby regulating the interfacial thermodynamic and hydrodynamic processes through three synergistic mechanisms. First, the micro/nano-scale asperities induce localized turbulence within the vapor layer, which enhances momentum exchange at the solid-vapor interface. Simultaneously, the recoil forces generated by these asperities disrupt the continuity of the vapor film, reducing interfacial thermal resistance[9,10,27,28]. Second, capillary-driven liquid spreading maintains intimate liquid-solid contact and effectively suppresses the formation of a continuous insulating vapor layer, ensuring robust thermal coupling while providing interconnected pathways for vapor evacuation[15,27,28]. Third, the hierarchical porous architecture facilitates boundary-slip–enhanced vapor transport, which significantly lowers viscous resistance and enables rapid radial drainage of vapor beneath the droplet (Fig. 1f)[15,27,28].

Furthermore, to gain a comprehensive understanding of the heat-transfer characteristics across different surfaces, we systematically investigated the constitutive relationship between the evaporation lifetime τ and the surface temperature T_w. As shown in Fig. 1g, the bare metal surface shows a typical two-phase heat transfer behavior: as T_w rises from 100 °C to 200 °C, τ decreases from 50.2 s to 1.644 s (a 96.7% reduction) due to enhanced thermal convection and increased Nusselt number. Once T_w exceeds the critical transition temperature (LTF of bare metal surface is 250 °C), however, τ sharply increases to 71.92 s (about 43.8 times higher), corresponding to the formation of a stable Leidenfrost vapor film (with a thickness >500 µm, measured via high-speed imaging). Remarkably, the NAVP surface maintains ultra-low τ values (0.417–5.124 s) across a broad temperature range of 100–1000 °C, which is lower than that of the bare metal surface, and no CHT transition point is observed.

Moreover, the regulation of aerogel density (D) is crucial for enabling efficient evacuation of the vapor generated in substantial quantities. Notably, the density D can be readily tuned by altering the concentrations of HCNFs and ceramic sol, allowing D to be varied over the range of 4–12 mg cm^{-3}[19]. As shown in Fig. 1h, we observe a non-monotonic dependence of droplet life time τ on material density D, reflecting the distinct failure modes at both low- and high-density limits. Specifically, at low densities (D < 7 mg cm^{-3}), the droplet exhibits a short τ (with an average of 0.519 s), primarily due to mechanical collapse of the nanofiber scaffold rather than efficient cooling. Cracks formed during impact accelerate local phase change and rapid evaporation, yet the overall structural failure of NAVP renders it unsuitable for practical applications. Conversely, at high densities (D > 9 mg cm^{-3}), excessive fiber packing leads to pore clogging, delaying vapor evacuation and promoting the reformation of a continuous vapor film, thereby prolonging τ (averaging 2.532 s) and reducing the heat flux. These two failure modes collectively define a well-defined bounded operational regime for NAVP functionality, with the optimal density window (~7–8 mg cm^{-3}) enabling complete droplet spreading, stable contact, and rapid phase-change cooling, with τ around 0.4 s at 1000 °C.

We further investigated the evolutionary pattern of the heat flux (q). As shown in Fig. 1i, the q exhibits a distinct stage-by-stage variation with increasing surface temperature (T_w). In the sub-Leidenfrost region (T_w < 250 °C), both the bare metal and NAVP systems display a similar

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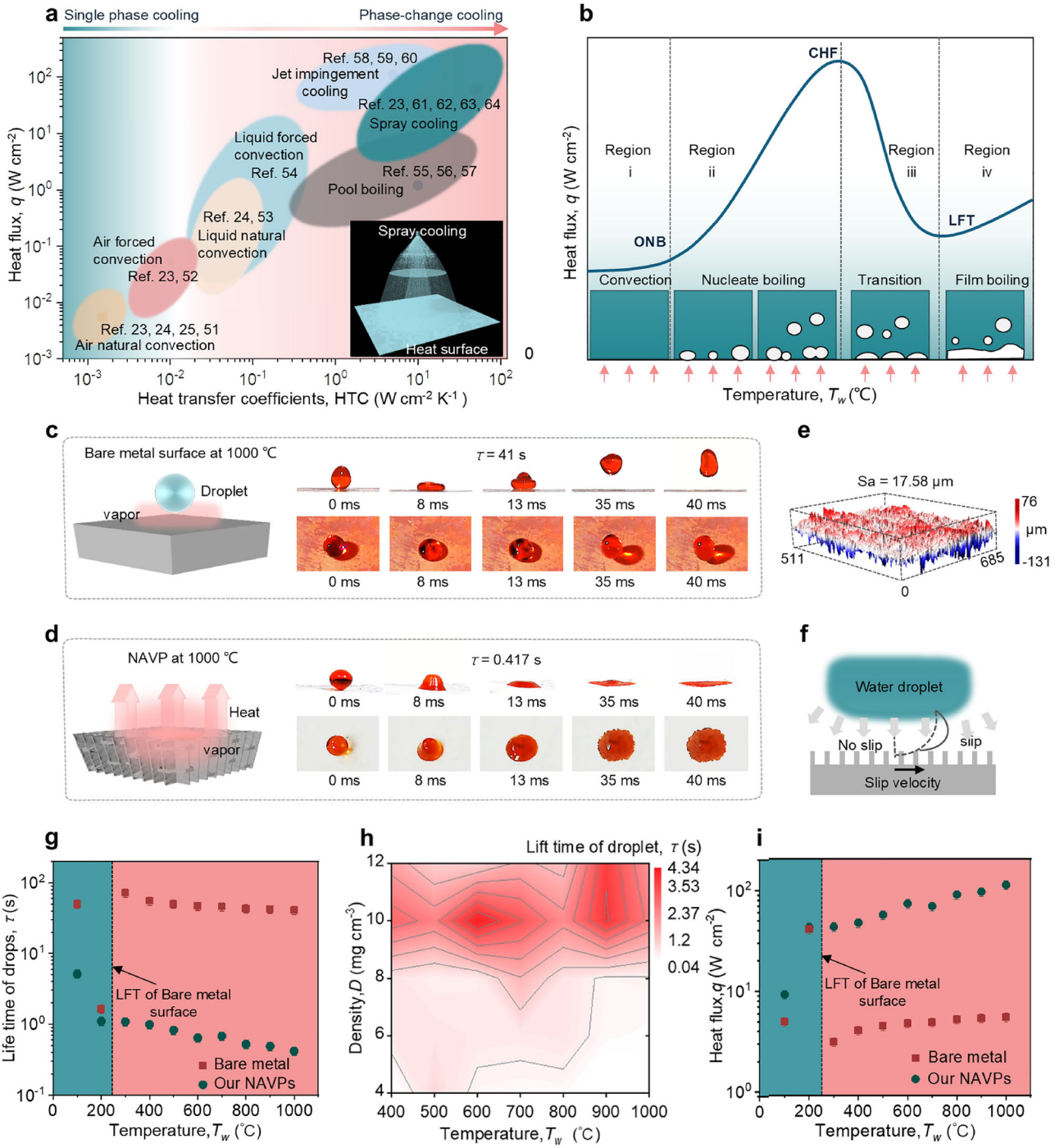


Fig. 1 | Design and principle of the NAVPs. a Comparison of different heat transfer methods, covering Air natural convection[23–25,51], Air forced convection[23,52], Liquid natural convection[24,53], Liquid forced convection[54], Pool boiling[55–57], Jet impingement cooling[58–60], and Spray cooling[23,61–64]. Color shading is used to distinguish cooling regimes: green indicates single-phase cooling, and red indicates phase-change cooling. **b** Schematic diagrams of typical phase-change cooling heat-transfer curves. ONB, the onset of boiling nucleating; CHF, the critical heat flux, is the local maximum in the heat flux, beyond which a much less effective film-wise mode develops; LFT, Leidenfrost temperature. **c** Left, schematic of a water droplet on a superheated bare metal surface, the generated vapor layer beneath the droplet cannot fully escape, forming an insulating barrier (the Leidenfrost effect occurs). Right, high-speed oblique and side views of drops impacting the bare metal surface at T_w = 1000 °C. **d** Left, schematic illustration of a water droplet on the NAVP surface, which consists of an array of microchannels that wick the incoming liquid and rapidly evacuate the generated vapor. Right, high-speed oblique and side views of water drops contacting NAVP at T_w = 1000 °C. **e** 3D surface profile of NAVP. **f** Diagram of a floating droplet at the moment of rebounding. **g** Lifetime of a water drop on bare metal surface (τ_{metal}) and NAVP (τ_{NAVP}), plotted as a function of the temperature T_w. The green-shaded region corresponds to the low-temperature regime (T_w < LFT$_{metal}$), while the red-shaded region represents the high-temperature regime (T_w > LFT$_{metal}$). **h** The phase diagram of the NAVP, in the form of a graph where the variables are the T_w and the aerogel density D. **i** The heat flux q as a function of substrate temperature (T_w). The values of q were calculated according to the procedure described in Supplementary Note 1. Source data are provided as a Source Data file.

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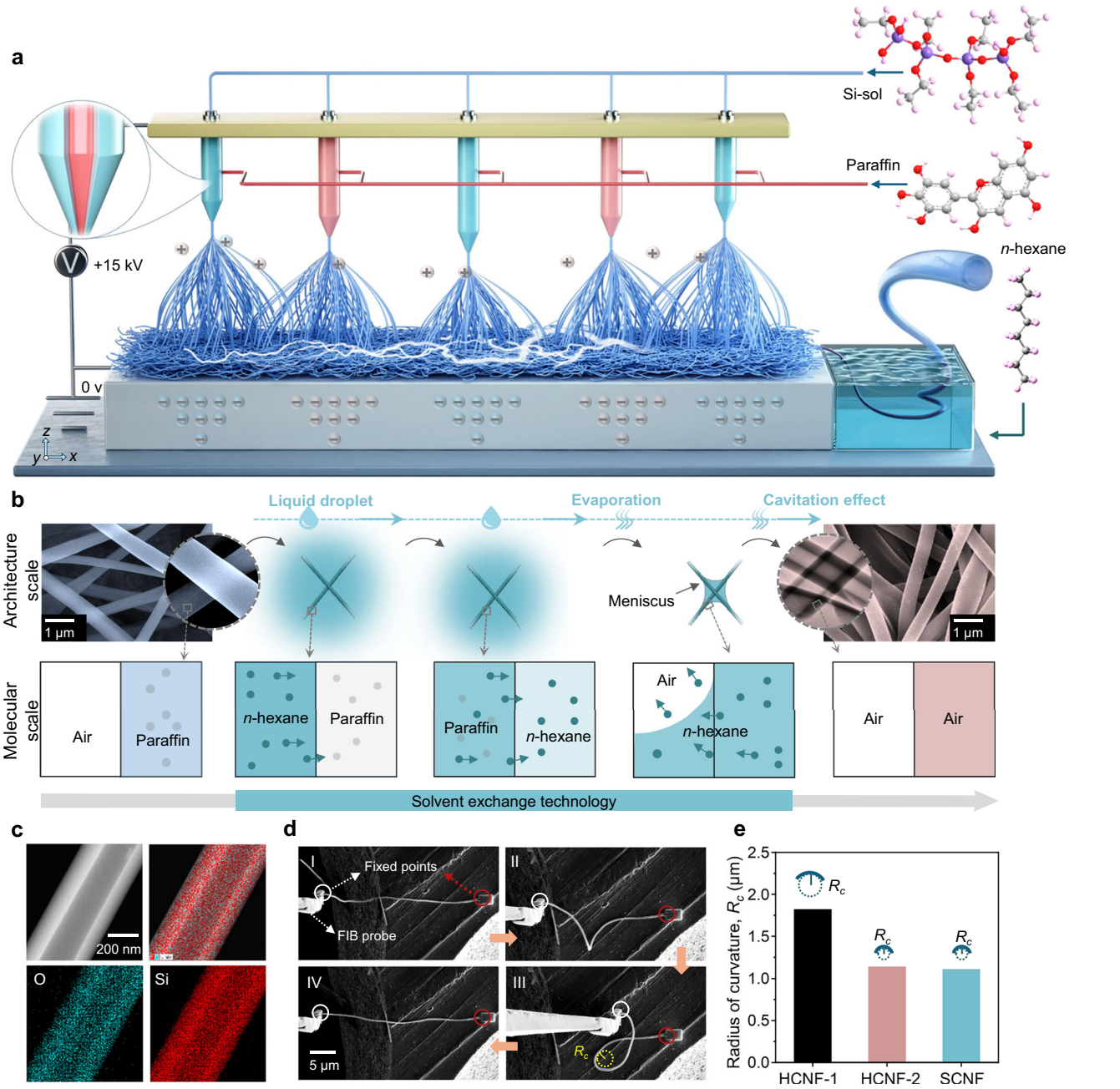


Fig. 2 | Formation of the HCNFs. a Schematic of the design and preparation procedure of the hollow ceramic nanofibers (HCNFs). Two immiscible spinning fluids are injected through concentric electrified needles, positioned at a short distance (10–12 cm) from a grounded collector electrode. A composite Taylor cone is generated, from which coaxial nanojets are ejected. Upon solidification of the outer fluid, core liquid-filled ceramic nanofibers are obtained. **b** Schematic illustration of the solvent exchange strategy prior to the evaporation of the core solution. **c** HRTEM-EDS images of an individual HCNF. **d** Top view of the in-situ bending test process: a HCNF was fixed at both ends to a substrate and then bent to determine the minimum bending radius (R_c). **e** Comparison of R_c values obtained for two individual HCNF (labeled HCNF-1 and HCNF-2, tested twice) and solid SiO_2 ceramic nanofibers (SCNF). Insets: schematic illustration to clarify the definition of R_c. Source data are provided as a Source Data file.

linear enhancement in q. However, once T_w exceeds the LFT of the metal (i.e. $T_w \geq 250$ °C), a pronounced divergence emerges between the two samples. Concretely, the NAVP system maintains an average heat flux of 72.22 W cm^{-2} across the supercritical temperature range (300–1000 °C), which attributed to the unique capillary structure and the associated specific surface area multiplication effect. In contrast, the metallic system experiences a sharp decline in q from 40.024 W cm^{-2} at 200 °C to 3.037 W cm^{-2} at 300 °C —nearly an order of magnitude reduction— attributable to the continuous formation of an insulating vapor film with a low thermal conductivity of 0.026 W m^{-1} K^{-1}.

Formation of the HCNFs

The NAVPs manufacturing process began with the formation of flexible, SiO_2 HCNFs by combing coaxial electrospinning[29], low-surface-energy solvent exchange technology, and a precisely controlled calcination process[30,31]. As illustrated in Fig. 2a, two injection syringes containing distinct electrospinning fluids were connected to the coaxial spinneret, which were mounted on an electrospinning apparatus[31]. Notably, the selection of spinning fluids were tailored based on the desired functional properties of the final material[32]. In the context of developing a thermal cooling material, it was of paramount

importance to consider the flexibility, mechanical strength, and thermal stability of the sheath material. Additionally, the incorporation of an inert, immiscible (or poorly miscible) liquid as the core was essential for attaining the desired hollow structure. Such liquids serve as versatile and adaptive hollow nanofibers templates that obviate the need for molecular-level assembly.

In this work, a silica precursor (SP) sol was employed as the sheath spinning fluid, while paraffin oil (PO) was used as the core spinning fluid[33]. Mechanistically, the gelation of the SP sheath prior to the electrically driven breakup of the coaxial PO nano jet promoted the formation of a core–sheath architecture with a PO-filled core, which acting as a liquid template. This process was meticulously optimized by controlling the cross-sectional dimensions of the sheath and core spinnerets and applying distinct feeding rates for the SP sol and PO solutions to ensure a uniform flow of both components. Under these optimized conditions, precursor fibers were successfully fabricated[29,31–33]. A metallic plate, serving as the counter-electrode or collector, was positioned at a short distance—typically several centimeters—from the spinning nozzle.

Essentially, upon the evaporation of the core solution in drying process, the resulting nanofiber structure has undergone a morphological transition from a liquid-filled template to air-filled nanofibrous architecture, culminating in the desired hollow form. However, HCNFs often experienced substantial deformations during the evaporation process, which was attributable to capillarity-produced pressure within nanoscale pores. A crucial step, therefore, is the precise exchange of the core PO liquid with low-surface-energy solvents before drying (Fig. 2b).

The evaporation of liquid contained in porous materials is typically governed via the following three distinct process: the liquid-vapor interface recession, the shrinkage due to drying, and the cavitation effect (Supplementary Fig. 4)[34]. In our work, the cavitation effect emerges as a pivotal factor in the drying process of core solutions (standard atmospheric pressure), beyond the scope of the two commonly recognized evaporation processes[35]. Mechanistically, cavitation occurs when the liquid confined within nanoscale pores is subjected to a capillary-induced tensile stress that exceeds its cohesive strength, triggering vapor-bubble nucleation within the gel matrix. This phenomenon is particularly pronounced when low–surface-energy solvents (e.g. *n*-hexane, *n*-octane) are involved, as the free energy barrier for cavitation scales strongly with the liquid's surface tension (γ^3) under constant pressure[34], making higher-γ liquids significantly more resistant to cavitation. Moreover, the inherent surface chemistry characteristics of silica ceramic network contribute to the cavitation process. The presence of surface defects, such as residual silanol groups from inadequate sialylation, facilitates heterogeneous nucleation of vapor/air bubbles. This process requires less energy compared to homogeneous nucleation, thereby promoting the preservation of the nanofibrous architecture[34,36,37]. Consequently, *n*-hexane was selected as the exchange bath to replace PO prior to the drying of the precursor fibers. In short, through cavitation-induced stress release, this approach efficiently mitigated drying stress and maintained the integrity of the hollow structure. In contrast, direct drying and calcination led to pronounced structural collapse (Supplementary Fig. 5), emphasizing the vital importance of solvent exchange and cavitation control in preserving structural integrity of the HCNFs.

Further morphological and structural characterization of the as-synthesized HCNFs were conducted using High-Resolution Transmission Electron Microscopy (HRTEM) imaging. Fig. 2b and Supplementary Fig. 6 show representative low-magnification HRTEM images of HCNFs, clearly revealing their hollow configuration. To determine the chemical composition of the HCNFs, energy-dispersive X-ray spectroscopy (HRTEM-EDS) mapping was performed on a single nanofiber. As shown in Fig. 2c, Si and O signals were homogeneously distributed along the hollow framework, confirming that the HCNFs are hollow nanofibers with amorphous Si-O-Si walls[38].

We further evaluated the mechanical flexibility of individual HCNFs by quantifying the radius of curvature (R_c) at the critical bending points, where extreme deformations occur. Particularly, one end of each HCNF was secured to a substrate, while the other was connected to a movable probe (Fig. 2d and Supplementary Fig. 7). This setup allowed the fiber to be bent until fracture, which facilitated the calculation of R_c[39,40].As illustrated in Fig. 2e, the R_c for HCNFs is determined to be ~1.82 μm (HCNF-1) and ~1.14 μm (HCNF-2), a value that aligns with that of solid SiO_2 ceramic nanofibers (SCNF) with a R_c of ~1.11 μm. The lower R_c value indicates enhanced flexibility, reflecting a more pronounced angular velocity of the tangential direction in relation to the arc[40]. This characteristic is pivotal for materials that are subjected to bending stress, as it implies a greater capacity for deformation without structural failure.

Fabrication of NAVPs

The fabrication of NAVPs was guided by two key design principles: (i) constructing a hierarchically interconnected HCNF aerogel with well-controlled tunneling orientation, and (ii) establishing a robust interfacial bonding network that integrates elastic deformability with thermal stability while preserving the integrity of the multiscale pore architecture. To fulfill the first principle, an oriented freezing-assisted molding approach was developed to precisely regulate the 3D assembly and directional alignment of HCNFs. To satisfy the second, a directed interfacial welding strategy was employed, wherein Methyltrimethoxysilane (MTMS)-derived silica sols (Supplementary Fig. 8) were introduced to modulate interfacial coupling and enable controlled bonding among HCNFs. This strategy features three key innovations: (i) the intrinsic chemical compatibility between the MTMS-derived silica precursor and the HCNF matrix ensures thermally stable interfacial adhesion; (ii) the tunable sol-gel transition kinetics enable molecular-scale gradient bonding across the fiber junctions; and (iii) the spatial domain-confinement effect reinforces interfacial connectivity without compromising the aerogel's framework topology or the hollow microstructure of HCNFs.

As presented in Fig. 3a, b, a straightforward method that integrated directional freezing-induced assembly and confinement targeted welding was developed to fabricate the NAVPs. Initially, short-chain *n*-alkane–filled (e.g., *n*-octane, *n*-hexane) HCNFs were selected to engineer targeted interfacial welding sites. The *n*-hexane–filled HCNFs were used as a model system, uniformly mixed with an MTMS-derived ceramic sol, and subsequently frozen directly in liquid nitrogen, enabling confined phase transition and localized interfacial welding during the subsequent sol-gel solidification process. As expected, a honeycomb-like micro network characterized by aligned pore walls was produced (Fig. 3h–l). To further increase the structural stability of the NAVPs, a high-temperature calcination process (in air) was performed. This procedure succeeded in removing the organic constituents within the NAVPs and facilitating transformed hydrolyzed ceramic sol (Supplementary Fig. 9) into a durable inorganic network, thereby strengthening the structural integrity of the NAVPs[19,40].

To gain molecular-level insight into the interfacial coupling mechanism, molecular dynamics (MD) simulations were subsequently conducted by shearing the MTMS sol confined between HCNF walls and *n*-hexane at the nanoscale (Supplementary Note 2)[41–43]. Specifically, the Lennard-Jones potential was employed to describe the interfacial interactions. A cut-off distance of 12.5 Å was applied, with van der Waals and Coulomb interactions calculated using atomistic and Ewald methods, respectively. After energy minimization, the sandwich structure was subjected to shear along the *z*-axis after 50 ps, 200 ps and 400 ps with a displacement rate of 0.05 Å ps^{-1} in the NVT ensemble regulated by Nosé thermostat. Periodic boundary

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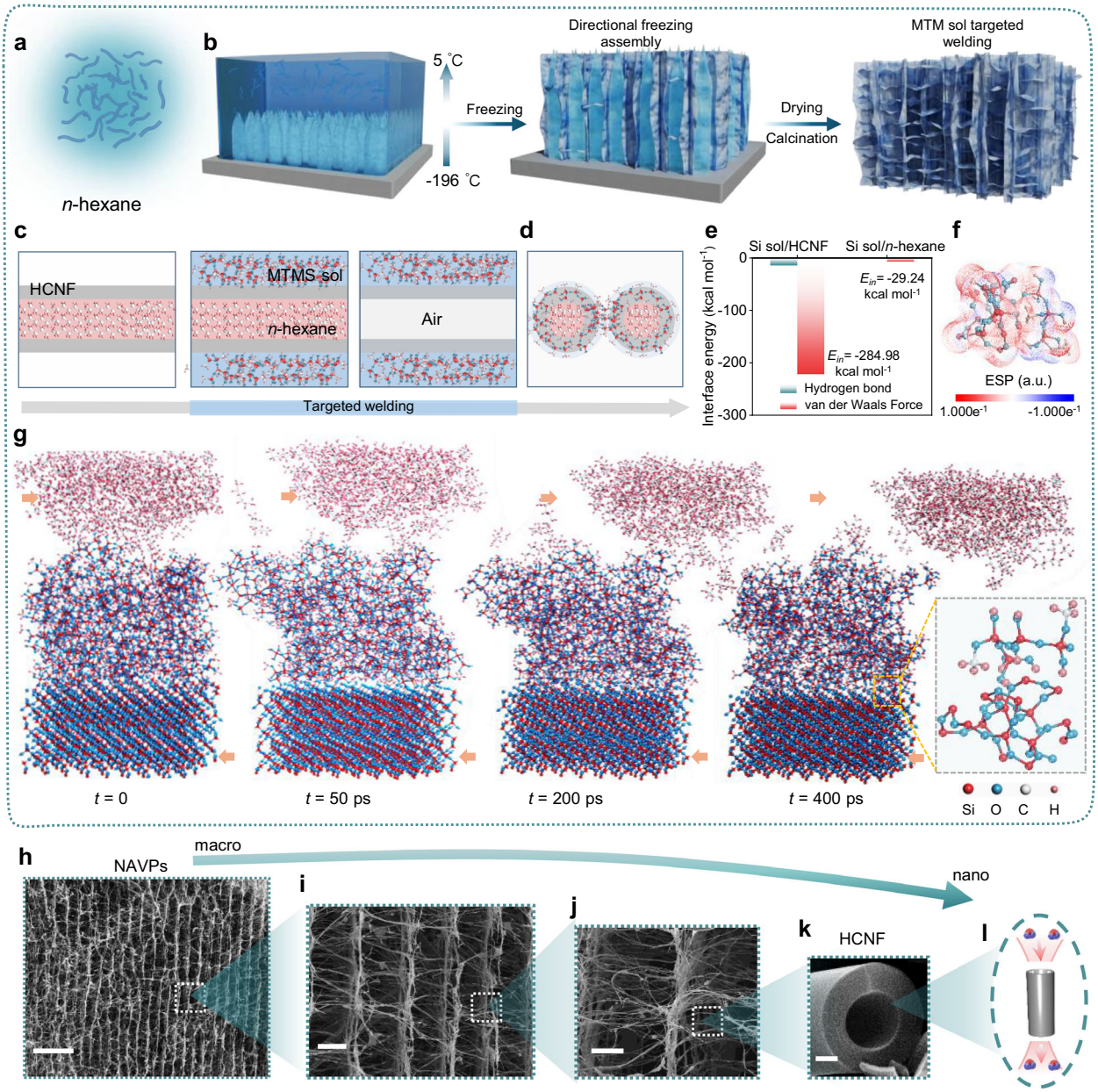


Fig. 3 | Fabrication of NAVPs. a Capillary infiltration of *n*-hexane into hollow ceramic nanofibers. **b** Schematic illustration of the fabrication of the aerogel-based vapor permeator (NAVPs) via directional freezing assembly and congener-targeted welding process. **c** Schematic of welding, that is, the tailored and confined interfacial coupling of hollow ceramic nanofibers (HCNFs) compounds through introducing Methyltrimethoxysilane (MTMS) silica ceramic sol to the *n*-hexane-filled HCNFs. **d** The schematic of the top view of *n*-hexane-filled HCNFs welded with silica sol. **e** Interface interaction energies between HCNF and silica sol. **f** The electrostatic potential (ESP) distribution of optimized silica sol/HCNF system is plotted from -0.1 to 0.1. **g** Simulated in-situ arrangement of the MTMS silica sol after 0 ps, 50 ps, 200 ps and 400 ps confined shear. **h**–**k** Scanning electron microscopy (SEM) images with different magnifications of NAVPs at various magnifications. Scale bars, **h** 100 μm; **i** 20 μm; **j** 20 μm; **k** 100 nm. **l** Schematic illustrates the process of vapor transport through a single HCNF. Source data are provided as a Source Data file.

conditions were alternately applied in two directions to minimize edge effects (Supplementary Fig. 10). The dynamic trajectory was recorded every 5 ps. As shown in the simulation snapshots (Fig. 3g), in sharp contrast to *n*-hexane, the ceramic sol exhibited a strong interfacial attraction to the HCNF walls, consequently forming a distinctive wall-induced layering configuration that is characteristic of confined sol–surface interactions[42,44]. The calculated interfacial interaction energy (E_{in}) for the Si sol/HCNF system reached −284.98 kcal mol^{-1}, which is nearly an order of magnitude more negative than that of the Si sol/*n*-hexane system (−29.24 kcal mol^{-1}), clearly indicating that the HCNF surface preferentially binds to the MTMS-derived Si sol rather than to *n*-hexane (Fig. 3e)[45]. This strong interfacial affinity can be primarily attributed to two dominant non-covalent interactions—hydrogen bonding (−15.34 kcal mol^{-1}) and van der Waals forces (−222.61 kcal mol^{-1})—while the electrostatic contribution remains negligibly small (Fig. 3e and Supplementary Table 1).

Moreover, the electrostatic potential (ESP) distribution of the Si sol/HCNF system further confirmed the presence of hydrogen bonding, as evidenced by the positive ESP regions (red) localized on hydrogen atoms and the negative ESP regions (blue) on oxygen atoms (Fig. 3f)[46]. This electrostatic attraction between oppositely charged sites promotes the formation of stable hydrogen bonds at the interface. Inversely, the nonpolar nature of *n*-hexane limits both physical adsorption and chemical bonding with the Si sol, consistent with the

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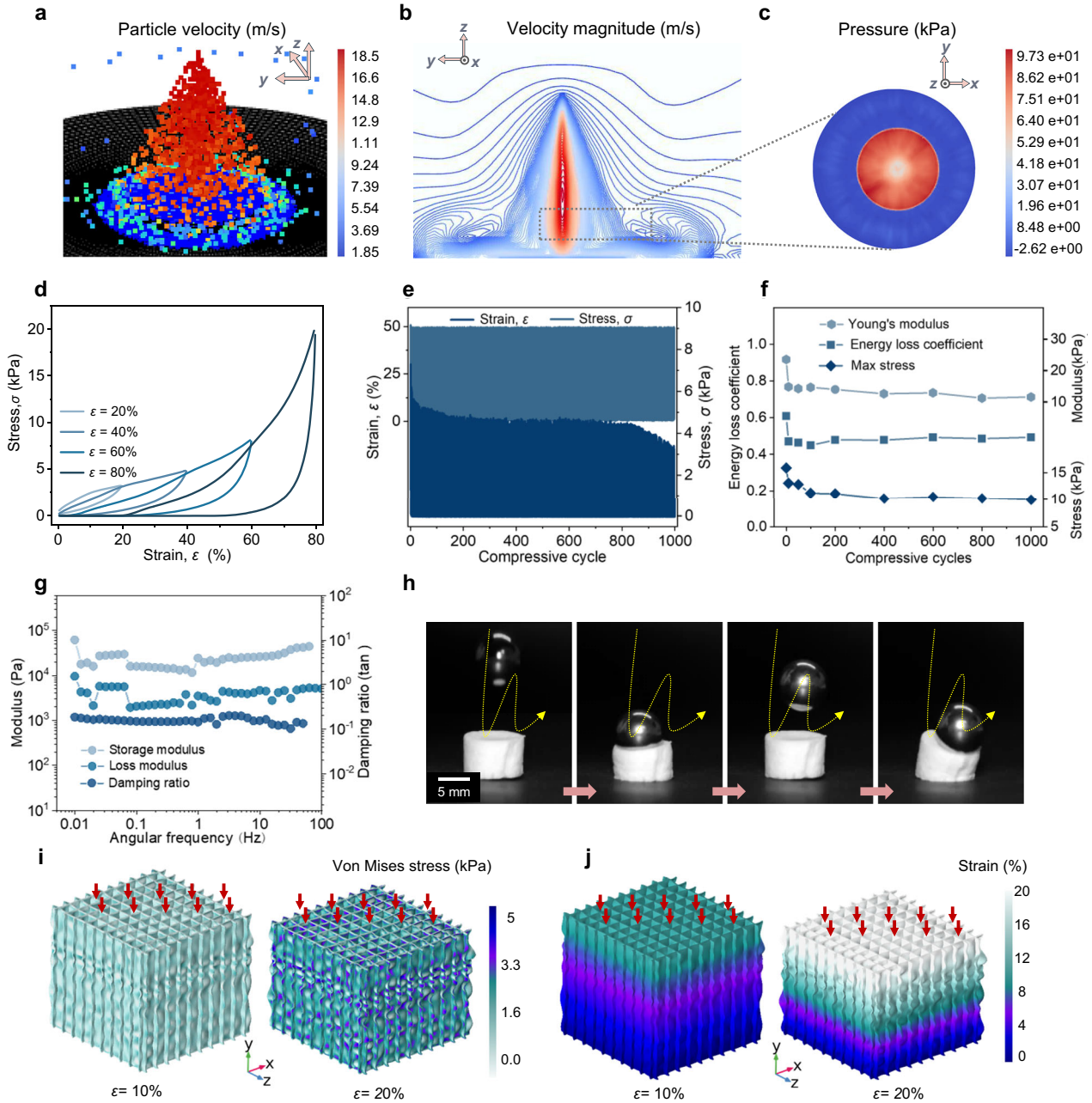


Fig. 4 | Mechanical properties of the NAVPs. Numerical simulations of **a** droplet velocity vectors, **b** velocity magnitude, and **c** pressure distribution during spray cooling. **d** Stress–strain (σ–ε) curves obtained from load–unload cycles with increasing strain amplitudes. **e** A fatigue test of 1000 compressive cycles at a strain amplitude of $\varepsilon = 50\%$. **f** Evolution of Young's modulus, energy loss coefficient, and maximum stress as a function of compressive cycles. **g** Frequency dependence of the damping ratio, storage modulus, and loss modulus of NAVPs at $\varepsilon = 3\%$. **h** Real-time images showing the rebound behavior of a high-speed steel ball on the NAVPs. Finite element simulations of **i** von Mises stress and **j** strain distributions within the NAVPs during impact. Source data are provided as a Source Data file.

weak interfacial adhesion observed in the simulation. In short, the stronger affinity of the silica sol for HCNFs, compared with *n*-hexane, provides a unique strategy for engineering anchored and confined welding sites. By filling the HCNFs hollow channels with nonpolar *n*-hexane, the inner spaces are effectively occupied, and the highly diluted concentration of the ceramic sol prevents excessive infiltration into the hollow channels and framework, thereby avoiding clogging or the formation of an uneven silica layer (Fig. 3c, d).

Encouraged by the above results, the NAVPs were further examined using SEM. As shown in Supplementary Fig. 11, the top-view SEM images reveal that the ceramic sol forms a continuous and uniform interconnecting layer among the HCNFs, effectively filling the interstitial regions after welding. Notably, no pore blockage was observed within the HCNFs, suggesting that the ceramic sol exhibits excellent chemical affinity and structural compatibility with the HCNFs[47]. These findings collectively demonstrate that the intrinsic hollow architecture of the HCNFs is well preserved throughout the NAVP fabrication process.

Mechanical property of the NAVP material

During spray cooling, the kinetic impact of atomized droplets imposes stringent requirements on the high-strain-rate mechanical properties of the cooling medium (Fig. 4a–c). To elucidate this impact resistance, the dynamic mechanical behavior of NAVPs were systematically

characterized via dynamic mechanical analysis (DMA). As shown in Fig. 4d, the NAVPs exhibit remarkable mechanical resilience, maintaining structural integrity under compressive strains (ε) as high as 80%. The corresponding stress (σ) reached 19.8 kPa, reflecting a load-bearing capability exceeding 10,000 times their own weight—an exceptional mechanical robustness rarely achieved in other porous materials. Furthermore, hysteresis curves from 1,000 loading–unloading compressive cycles at $\varepsilon = 50\%$ revealed negligible plastic deformation (Fig. 4e and Supplementary Fig. 12), and the NAVPs retained more than 50% of their maximum stress and initial Young's modulus after cycling (Fig. 4f), confirming their remarkable structural robustness and cyclic durability. Dynamic compressive tests conducted via DMA further revealed that both the storage modulus and loss modulus remained nearly constant and independent of angular frequency across three orders of magnitude (0.01–100 Hz) (Fig. 4g). The measured damping ratio (~0.15) over the entire frequency range indicates that the elastic response predominates in the NAVPs, which can be attributed to their robust and viscoelastic nanofibrous network. In addition, the dynamic deformation process of the NAVP under impact loading was captured by a imaging system. The real-time image sequence demonstrates that the NAVP can rebound a 6 mm diameter bearing steel ball twice (Fig. 4h), clearly confirming its exceptional dynamic elasticity.

The mechanical response of the NAVP was comprehensively investigated through multi-physics field-coupled finite element analysis (Supplementary Note 3). A 3D network model was developed and discretized using tetrahedral meshing, and vertical compression was simulated within the framework of a linear-elastic eigenstructure model. The simulation results reveal a pronounced structure-oriented stress distribution within the 20% strain range (Fig. 4i, j). Most of the applied stress is transmitted along the primary axis of the honeycomb tubular network, while the stress concentration factor (K_t) at the fiber junctions ranges from 1.15 to 1.28—significantly lower than that of conventional foam materials ($K_t > 2.5$). This enhanced stress homogenization capability arises from three synergistic mechanisms: (i) the bionic honeycomb architecture guides multi-direction load diversion; (ii) the confined welding strengthens inter-fiber coupling and prevents interfacial debonding, and (iii) the controlled elastic buckling within the hollow tubular cells dissipates strain energy reversibly. Consequently, the octahedral topology at HCNF cross-nodes stabilizes the stress anisotropy index (AI) within 1.8–2.3—far below that of disordered porous networks ($AI > 5$)—thereby ensuring homogeneous stress transfer and effectively mitigating brittle fracture[48,49].

Thermal cooling applications of the NAVP material

We finally carried out a systematic investigation into the performance challenges of spacecraft thermal protection systems (TPS) under extreme multi-physics field-coupled environments. As depicted in Fig. 5a, the spacecraft experiences cyclic temperature fluctuations ranging from −120 °C to 100 °C during orbital operation; in the jump re-entry orbit, moreover, the TPS is exposed to a time-sequential coupling of three extreme thermal load stages. (i) The process begins with an initial high-thermal-shock phase, characterized by a transient temperature surge exceeding 3000 °C with a duration of less than 120 s. (ii) This is subsequently followed by a low-temperature vacuum transition phase, during which the ascent to higher orbital altitude induces rapid surface cooling to −90 °C at a rate exceeding 500 °C s^{-1}. (iii) Finally, in the terminal extreme re-entry phase, the TPS encounters a secondary atmospheric entry, during which it is subjected to ultra-high-temperature plasma flow (≈1900 °C) accompanied by intense aerodynamic shear stress exceeding 12 kPa. The thermal-mechanical coupling boundary conditions have been rigorously validated through high-enthalpy wind-tunnel simulations and multi-scale fluid–structure interaction analyses (Fig. 5b and Supplementary Fig. 13).

To address the thermal management limitations of conventional cooling materials, we designed and constructed a multimodal extreme thermal environment experimental platform, integrating a gradient heat-flux loading module, a high-speed infrared thermography system, and an active spray cooling unit (Supplementary Note 4 and Supplementary Fig. 1). The gradient heat-flux module employs a 15-kW electromagnetic induction heater coupled with a water-cooling circuit, enabling precise control of heat-flux density within 300–1300 °C. Notably, for low-temperature experiments (100–200 °C), a precision-controlled heating stage was employed to maintain uniform thermal boundary conditions. The high-speed infrared camera continuously recorded real-time surface temperature distributions (Supplementary Fig. 14), providing synchronized thermal field data for transient analyses. Meanwhile, the active spray cooling unit enabled dynamic regulation of the coolant flow rate (0.5–9.5 mL min^{-1}) and impact frequency via pressure modulation, allowing accurate reproduction of the transient liquid–vapor interaction during re-entry.

As shown in Fig. 5c, droplets were continuously released at a flow rate (Q) of 0.5 mL min^{-1} from a vertical height of 6 mm, with the release interval carefully set to exceed the surface thermal relaxation time, ensuring that the temperature recovered to its initial equilibrium state between successive impacts[3]. The experimental results reveal that the temperature fluctuation on the bare metal surface after a single droplet impact is negligible, as the Leidenfrost effect induces a stable vapor cushion that prevents direct liquid-solid contact and severely limits transient heat exchange. In sharp contrast, the NAVP surface exhibits a temperature reduction of about 250 °C following a single droplet impact, driven by rapid liquid imbibition and efficient vapor evacuation through its multiscale porous network (Supplementary Fig. 15). This superwetting-mediated cooling behavior remains highly stable, with the cooling-efficiency decay rate below 20% even after 17 successive impact cycles, confirming the exceptional thermal durability and impact-resilient heat dissipation capability of the NAVP material.

Moreover, the NAVP material exhibits a cooling enhancement under repeated droplet impacts. As shown in Fig. 5d, e, at a flow rate of $Q = 4$ mL min^{-1} and an initial surface temperature of 1000 °C, the temperature drop (ΔT) increases to approximately 691 °C. This reduction reflects a cumulative cooling effect arising from sequential liquid replenishment, intensified capillary infiltration, and accelerated vapor evacuation within the hierarchical porous framework of the NAVP, thereby enabling efficient and sustainable heat dissipation even under extreme high-enthalpy conditions. To quantitatively elucidate this cumulative cooling enhancement, the time-dependent evolution of the interfacial heat transfer can be analyzed using a transient thermal resistance model (Eq. 1). The progressive reduction in R_{th} (t) originates from persistent thin-film evaporation and continuous improvement of interfacial heat transport at the droplet-substrate interface:

$$R_{th}(t) = \frac{1}{k_{eff}} \sqrt{\frac{\alpha_{eff} \cdot t}{\pi}} \cdot \frac{1}{1 + Bi(t)} \tag{1}$$

where the time-dependent Biot number is defined as:

$$Bi(t) = \frac{h_{eff}}{k_{eff}} \sqrt{\pi \alpha_{eff} \cdot t} \tag{2}$$

Here, k_{eff} and α_{eff} represent the effective thermal conductivity and thermal diffusivity of the porous NAVP substrate, reflecting internal heat conduction. The interfacial heat transfer coefficient h_{eff} accounts for phase-change-induced evaporation and vapor-phase transport at the droplet–surface interface. As repeated impacts increase the thermal penetration depth $L(t) = \sqrt{\alpha_{eff} \cdot t}$, the Biot number correspondingly rises, indicating a transition from conduction-dominated to

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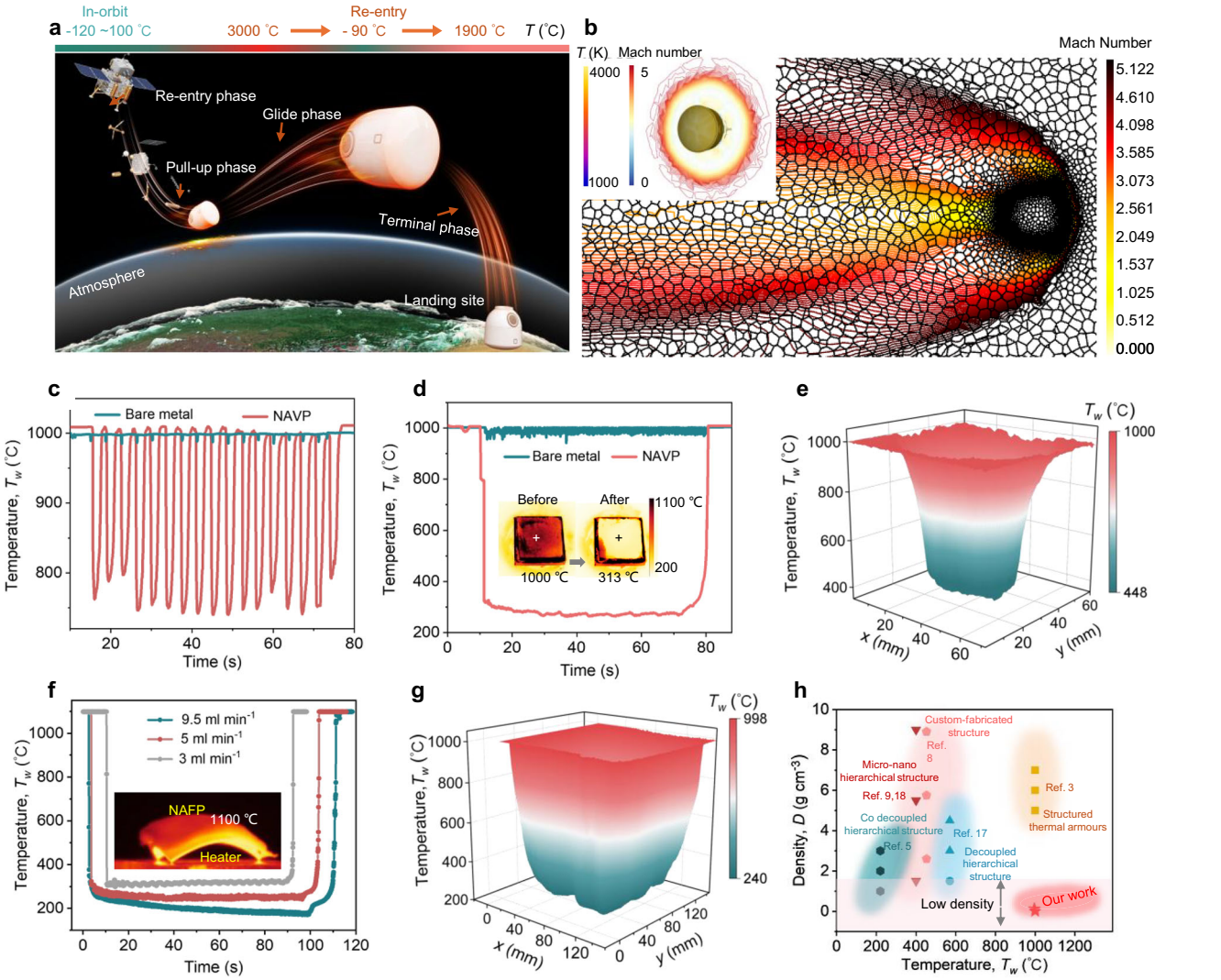


Fig. 5 | Thermal cooling applications of the NAVP material. a Schematic illustrating the temperature variations of the lunar probe during its orbital and re-entry phase. **b** Computational fluid dynamics (CFD) simulation based on the finite volume method showing the velocity distribution around the lunar probe re-entry capsule in a wind tunnel. Inset: coupled temperature–velocity fields obtained from the CFD simulation. **c** Temporal evolution of surface temperature for the bare metal and NAVP during droplet impacts at a flow rate of $Q = 0.5$ ml min^{-1}. **d** Temporal evolution of surface temperature for the bare metal and NAVP during droplet impacts at a flow rate of $Q = 4$ ml min^{-1}. Insets: optical and infrared images of NAVPs after exposure to water droplets at $Q = 4$ mL min^{-1} for 80 s. **e** Cumulative cooling effect of water droplets at $Q = 4$ mL min^{-1} for 80 s, reducing the surface temperature to 364 °C. x and y denote the length and width of the cooled NAVP surface, respectively. **f** Cooling performance of NAVPs under varying flow rates ($Q = 3$–9.5 mL min^{-1}): with a larger Q, a faster cooling response, thereby leading to a lower temperature at which NAVPs are cooled. Inset: photograph of a curved NAVP sample, demonstrating its excellent shape adaptability and mechanical flexibility. **g** Cumulative cooling under a flow rate of $Q = 9.5$ mL min^{-1} for 80 s, reducing the surface temperature to ~250 °C. x and y denote the length and width of the cooled NAVP surface, respectively. **h** Comparison of density and maximum working temperature (i.e., Leidenfrost temperature, LFT) among various structure materials, including Decoupled hierarchical structure[17], Co decoupled hierarchical structure[5], Structured thermal armours[3], Micro-nano hierarchical structure[9,18], Custom-fabricated structure[8]. Source data are provided as a Source Data file.

interface-limited heat transfer. This dynamic shift produces a nonlinear yet asymptotic reduction in R_{th} (t), thereby maintaining enhanced and self-sustaining cooling performance. In addition, the nanoscale surface features of the NAVP promote asymmetric contact-line pinning, effectively suppressing the Leidenfrost effect and preserving efficient heat exchange even at ultrahigh temperatures.

Flow-dependent studies further underscore the active thermal management capability of NAVP system (Fig. 5f, g). Increasing the flow rate to $Q = 9.5$ mL min^{-1}, corresponding to the jet cooling mode, the surface temperature undergoes an exponential decay from 1000 °C to 240 °C ($\Delta T = 760$ °C) over a period of 30 s (Supplementary Fig. 16). Such a rapid and monotonic temperature drop demonstrates that the NAVP responds sensitively to both flow rate and thermal load, exhibiting a quasi-linear flow-temperature coupling behavior that enables adaptive regulation under dynamic heat flux conditions. More strikingly, NAVP demonstrates the synergistic advantages of ultralightweight properties and high LFT compared to conventional active cooling materials (Fig. 5h and Supplementary Table 2)[3,17,18,47,48], offering a promising potential for spacecraft re-entry thermal protection and the heat dissipation of high-power density electronics.

Discussion

In summary, a compositional-structural engineering strategy was proposed, which integrates solution exchange in the nanofiber preparation process with targeted inter-fiber welding of the resulting aerogel, achieving the synergistic construction of NAVP. Remarkably, leveraging low–surface-energy solvent cavitation to relieve drying stress and confined welding to enhance interfacial coupling yields

NAVPs with hierarchically integrated tunnel architectures, enabling efficient vapor evacuation under Leidenfrost conditions. Importantly, the developed NAVPs effectively suppress the Leidenfrost effect even at temperatures approaching 1000 °C, while maintaining an ultralow density of 7 mg cm^{-3}. The proposed percolator provides a potential approach to mitigate the Leidenfrost effect for efficient heat transfer.

Methods

Materials

Tetraethyl orthosilicate (TEOS, Si $(OC_2H_5)_4$, 98%, Greagent), Methyltrimethoxysilane (MTMS, Shanghai energy chemical Co. Ltd.), Phosphoric acid (H_3PO_4, 85 wt%, Aladdin), Oxalic acid ($H_2C_2O_4$, 99%, Aladdin), *n*-hexane (C_6H_{14}, >99%, Greagent), Paraffin oil (CP, Greagent), Ethanol (EtOH, 99.7%, Greagent), Poly (vinyl alcohol) 1788 (PVA1788, Aladdin). Deionized water (resistivity > 18.2 MΩ cm) was used throughout all experiments.

Experimental methods

Preparation of silica precursor sol. The silica precursor sol was synthesized via the sol-gel method[50]. In a typical procedure, TEOS was subjected to hydrolyzed in the presence of H_3PO_4 as a catalyst for 8–10 hours, maintaining a mass ratio of MTMS: H_2O: H_3PO_4 at 100:100:1.1. After complete hydrolysis, a 13–14 wt% aqueous solution of PVA1788 was introduced into the silica sol at a mass ratio of 1:1. The resulting mixture was further stirred for an additional 3 h at ambient temperature to ensure full homogenization, yielding the SiO_2 precursor (SP) sol.

Preparation of HCNFs. Hollow SiO_2 ceramic nanofibers (HCNFs) were fabricated via a synergistic approach combining coaxial electrospinning and solvent exchange technology, followed by a calcination process. Initially, SP sol and paraffin oil (PO) were simultaneously processed using a coaxial electrospinning setup at a voltage of 14 kV. The collection distance was 12-15 cm, while flow rates for sheath/core fluids were 1 mL h^{-1}/ 0.2 mL h^{-1}. The operations were conducted under controlled environmental conditions (humidity 28 ± 3%; temperature 25 ± 3 °C). Subsequently, the as-spun nanofibers were immersed in *n*-hexane for 1–4 hours to enable sufficient diffusion and complete replacement of the PO phase by *n*-hexane, producing *h*-nanofibers. Finally, the *h*-nanofibers were air-dried at room temperature for 12 h and then calcined at 800 °C for 1 h to obtain HCNFs with well-defined hollow structure.

Preparation of NAVPs. NAVPs were fabricated via a directional freeze-drying method. Specifically, 0.4 g of MTMS with 0.003 g of oxalic acid were dissolved into 80 g water and stirred for at least 40 min to prepare MTMS silica ceramic sol[42]. Furtherly, 0.14 g of solid SiO_2 ceramic nanofibers (SCNF) (Supporting Methods) were introduced to the 40 g MTMS sol and uniformly dispersed to yield SCNF/MTMS dispersion. To achieve NAVPs with a target density (7 mg cm^{-3}), another 0.14 g of HCNFs were uniformly dispersed in *n*-hexane and stirred for 5 h to allow the solvent to penetrate the hollow channels of HCNFs, producing *h*-HCNFs. The *h*-HCNFs were then incorporated into the SCNF/MTMS dispersion, and the resulting mixture was rapidly poured into a directional mold, followed by freezing at −120 °C for 5 min in a liquid nitrogen bath. The frozen monoliths were freeze-dried for 36 h in a vacuum freezer to remove ice crystals and maintain the directional porous framework, forming dried aerogels. Subsequently, the dried aerogels were calcined at 900 °C to accomplish the inorganic conversion and enhance the structural stability, producing NAVPs.

Characterization

The microstructures of NAVPs were characterized by FE-SEM (S-4800) and TEM (JEM-2100F). Compression mechanical tests were performed using a DMA (TA-Q850) instrument. The fast recovery of NAVPs rebounding from a high-speed steel ball was recorded by a Fuji Film X-T200 camera operated at 800 frames per second. The experimental set-up of spreading and evaporation dynamics of water drops mainly consisted of an induction heater, a syringe pump (LM60B-RZ1030-4), two high-speed cameras, and an infrared camera. The induction heater (PIEN-10GT-3SZ) was used to maintain the samples at a prescribed temperature, in the range of 300–1150 °C. The precision-controlled heating stage (model JF-956) used for experiments in the low-temperature range (room temperature to 250 °C). The oblique and side views of water drops were filmed with two high-speed cameras NAC MEMRECAM HX-7S (2560*1680), NAC MEMRECAM GO-5M (2560*1920) operating at 1000 frames per second.

Data availability

Source data are provided as a Source Data file. Source data are provided with this paper.

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Author contributions

Y.S. and H.W. conceived and designed the study. H.W., Y.X., P. L. and Q. L. organized the data. H.W. wrote the manuscript. L.C. and J.Y. supervised the project. All authors discussed the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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